



Design of an energy-efficient XOR gate in QCA with applications in reversible logic-based one-bit comparator and ALU

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Abstract

A Quantum Dot Cellular Automata (QCA) is a nanotechnology-driven computing method that leverages quantum mechanical principles. This work demonstrates the efficient use of reversible logic gates in QCA-based systems, a crucial aspect of scalable and energy-efficient computer designs. In this research, we propose an energy-efficient, compact XOR gate and several common reversible logic gates. The paper further illustrates the use of these gates to construct essential components, such as a one-bit comparator and an arithmetic logic unit (ALU). The ALU, designed with Feynman and Toffoli gates, is capable of performing eight arithmetic and logical operations. We conduct a comprehensive evaluation of cell complexity, area efficiency, delay, area-delay product (ADP), and energy dissipation. Additionally, we compare the characteristics of the proposed circuits with prior efforts to highlight advancements and identify areas for further improvement in reversible computing paradigms. The recommended architecture enhances the performance of the XOR gate, reversible logic gates, one-bit comparator, and ALU. The XOR gate achieves a 67.86% reduction in cell complexity, an 85% improvement in area efficiency, and a 50% reduction in quantum cost. Feynman and Toffoli gates demonstrate a 75% reduction in area and an 87.5% reduction in quantum cost. For the one-bit comparator and ALU, the proposed solution reduces area by 87% and latency by 40%, saving both space and time. With a quantum cost that is 90% lower than traditional designs, the proposed architecture optimizes quantum circuits for real-world applications.

Keywords QCA · Reversible logic · ALU · Energy analysis · QCADesigner-E

1 Introduction

As device dimensions near quantum limits, conventional scaling encounters major challenges due to fundamental effects, requiring a change from the FET-based paradigm. CMOS devices have basically reached their intrinsic limits,

rendering any additional reduction in feature size undesirable to optimal operation [1]. This challenge influences the investigation of alternate device technologies, one of which is quantum dot cellular automata (QCA). QCA shows up as a possible approach enabling high-frequency operation, low-power consumption, and high-density devices [2–4].

QCA operates on the principle of electron confinement and mutual repulsion, distinguishing it from conventional transistor-based circuits. Unlike transistors, QCA avoids the discharging of capacitors during logic value transitions eliminating energy dispersion. Coulombic interaction drives polarization propagation between cells, transferring information without the need for current flow. Each QCA cell restricts a defined number of charges in specific locations, allowing tunneling between them. Coulomb effects play a dominant role, fostering interactions between neighboring cells [5, 6].

Reversible logic offers a solution to the energy dispersion problem associated with irreversible logic. By minimizing information loss, it enables more energy-efficient computing,

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particularly vital in the era of power-conscious electronics [7]. The principles of reversible logic have led to the development of numerous reversible logic gates. Reversible logic gates serve as a basic component for sustainable computing, whose primary focus is reducing the consumption of energy and creating a null impact on the environment.

The arithmetic logical unit (ALU) acts as the basic functional module within central processing units (CPUs), handling logical and arithmetical operations. Logic gates are fundamental in the ALU's design and operation. ALUs with reversible logic gates use less power, which increases the benefits of QCA technology, such as nanoscale circuits and terahertz performance.

In this paper, we concentrate on the implementation of an energy-efficient XOR gate and its application in the development of basic reversible logic gates. These gates are efficient and closely linked to circuit complexity, area, delay, and quantum cost when implemented in QCADesigner E. It extends its investigation by integrating these reversible gates into circuit designs, including a one-bit comparator and an arithmetic logic unit (ALU), aiming to assess their performance and applicability. This study seeks to provide valuable insights into the feasibility and advantages of reversible logic gates in QCA-based circuitry. The following structure guides the subsequent sections of the paper: Section 2 briefs the literature survey. Section 3 provides an overview of the fundamentals of QCA. Section 4 elaborates on the proposed reversible gates, simulation results, and comparisons. Section 5 presents the simulation and comparison of the proposed one-bit comparator. Section 6 focuses on the proposed ALU and its simulation and comparison. Finally, Section 7 encapsulates the study's findings and explores prospective paths for future research.

2 Literature survey

This review of the literature looks at ways to make QCA circuits work better by looking into reversible logic design, advanced ALU designs, energy-efficient circuits, and simulation and validation methods. In [8], Feynman and Toffoli gates were suggested as a way to reduce complexity. In [9], the Layered-T method was introduced as a way to reduce cell count, but it has many problems. In [10], authors designed an ALU that includes coplanar crossings that make possible many operations and preserve efficiency. Amidst this, authors [11] and [12] introduced multilayer crossover designs that balance complexity and flexibility for QCA, showing their applicability in various processes.

The authors [13] presented a novel 3-input XOR gate designed for minimal complexity and maximum performance, functioning as the basis for the development of an efficient QCA one-bit full adder and a 4-bit Ripple Carry

Adder (RCA). These designs demonstrated improved efficiency and robustness, achieving significant improvements in area, cell count, latency, and cost compared to existing implementations.

The authors [14] designed fault-tolerant three-input majority gates for multiplexers. The authors [15] introduced a cost-efficient and scalable architecture of a QCA-based ALU using reversible TSG and AND–OR gates, with FPGA-based validation proving its low-power consumption, improved fault tolerance, and robust performance. In [16], authors suggested a QCA-based full adder utilizing XNOR gates to improve performance and minimize the power consumption. Also, [17] showed a powerful QCA-based Hamming code generator and error correction using Feynman gate, and complexity tests proved how well it worked. These studies emphasize QCA's capability in creating scalable and energy-efficient circuits.

In [18], the authors presented a very efficient reversible Full adder with three-input and five-input majority gates. Thermal robustness and coplanar crossing optimizations guaranteed reliable design performance, which were subsequently enhanced to develop ripple-carry adders with increased efficiency. Focusing on better QCA parts highlights improvements in small size, area efficiency, and lower quantum cost, which makes it easier to create useful QCA circuits.

Simulation and validation techniques are crucial for QCA designs. The authors of [18] proposed an innovative eigenstate computation approach for Hamiltonian basis sets, enhancing accurate simulations and error identification in prior designs. These methods use QCA Designer modeling tools like those in [17] and [18] to demonstrate the importance of precise validation procedures in ensuring that QCA-based systems are reliable and resilient.

Furthermore, in [19] authors outlined a multilayer reversible arithmetic logic unit (RALU) in QCA, including XOR, OR, AND, and adder circuits to enhance cell count, spatial area, and latency. The work in [20] examined the creation of a 20-operation ALU employing reversible logic and QCA technology, incorporating Feynman and Fredkin gates to minimize energy dissipation. The authors of [21] came up with a new way to compute eigen states for QCA device simulations using Hamiltonian modeling. This made it easier to find problems in older designs.

In [22], the authors explored advances in unconventional computing paradigms, including nanotechnology, quantum, DNA, and residue-based systems and their important roles in deep learning model and post-quantum cryptography methods, while in [23–25], the authors highlight the techniques for improving circuit efficiency, energy optimization, and reliability in QCA and reversible computing.

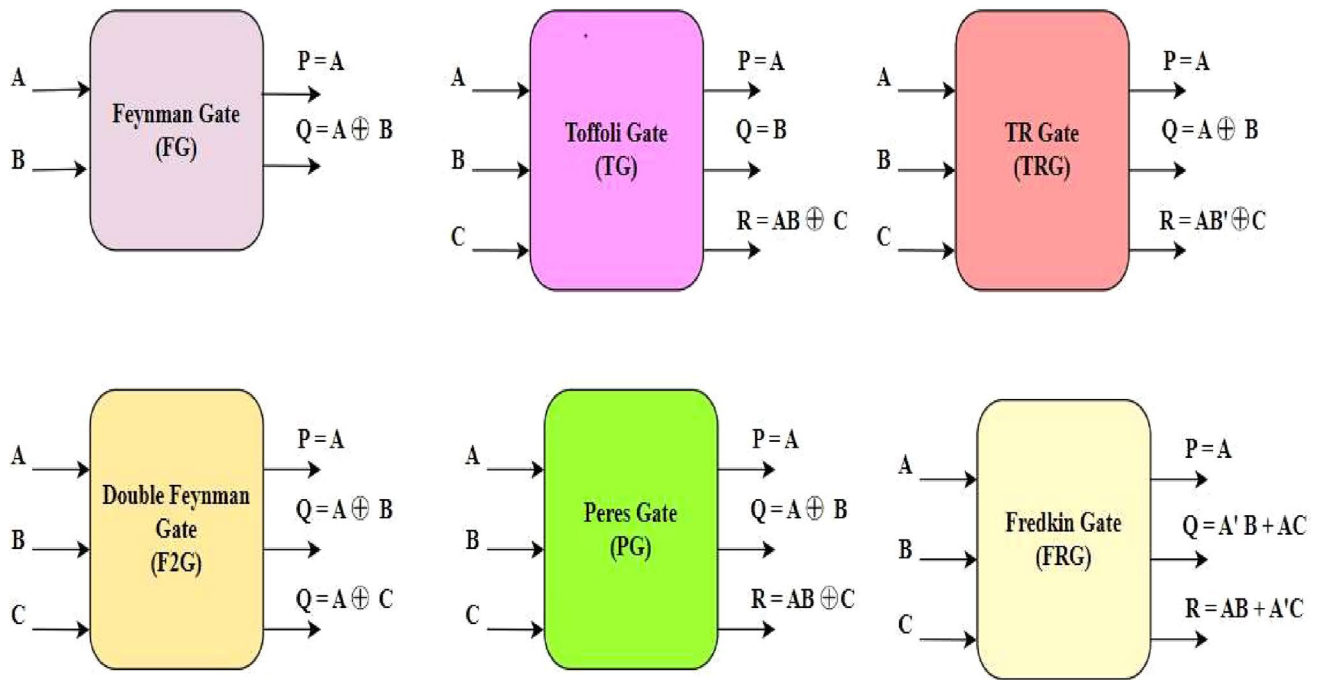


Fig. 1 Different types of reversible logic gates

3 Quantum dot cellular automata: a brief overview

3.1 Quantum cell

The QCA cell is the basic building block of QCA systems. It consists of four quantum dots arranged in a square with two excess electrons. Only two stable states exist where electrons occupy opposite diagonal sites, representing binary “0” and “1.” Electron placement follows Coulombic repulsion and tunneling effects. QCA cells are mainly of two types: 90° and 45° cells [7].

3.2 QCA clocking

Information transfer occurs through polarization changes between adjacent cells. Clocking divides cells into four zones (0–3), each with phases Switch, Hold, Release, and Relax. A 90° phase shift ensures unidirectional flow and synchronization. The clock also provides power by modulating tunneling barriers. Clocking approaches include Landauer and Bennett mechanisms [8].

3.3 QCA wiring and crossovers

Binary wires are formed using aligned 90° or 45° cells. Crossovers are required for circuit interconnections. Methods include multilayer (accurate but costly), coplanar (area-efficient but

prone to misalignment), and clock zone-based (reliable and compact) [26].

3.4 Fundamental gates

Basic logic in QCA is built using Majority Voter (MV) and Inverter gates. An MV gate outputs the majority of three inputs:

$$\text{Maj}(X, Y, Z) = XY + YZ + ZX$$

AND/OR gates are derived by fixing one input as 0 or 1. An inverter uses two cells with opposite polarization [27].

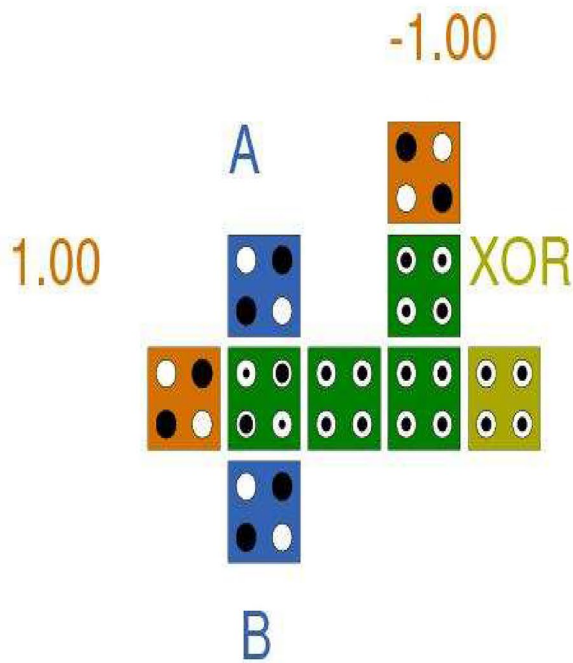
3.5 Kink energy

Kink energy is the energy difference between adjacent cells with opposite and same polarization. Lower kink energy improves stability and switching reliability [28–30]. Mathematically, the Kink Energy (E_k) is expressed as:

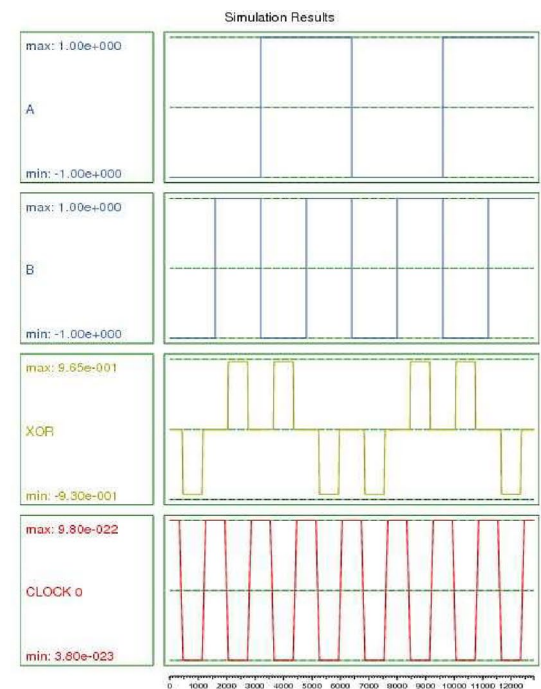
$$E_k = E_{\text{opposite}} - E_{\text{same}}$$

3.6 Energy and power dispersion

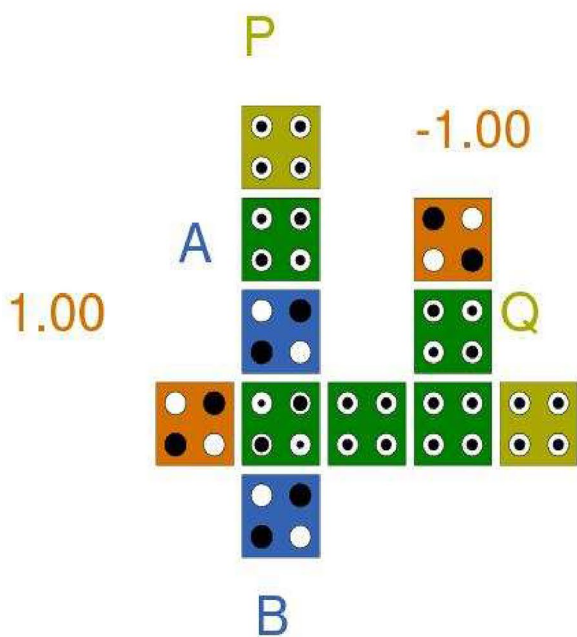
Polarization under clocking causes energy transfer between cells. As specified in [8], cell energy:



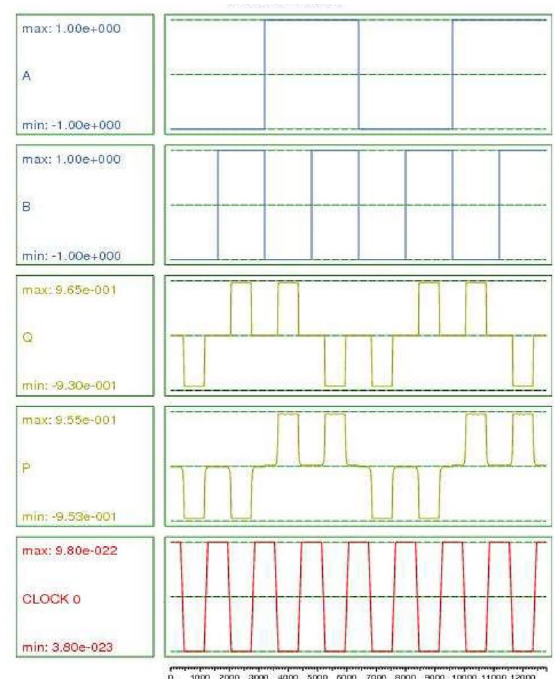
(a) QCA XOR gate



(b) XOR gate output waveform

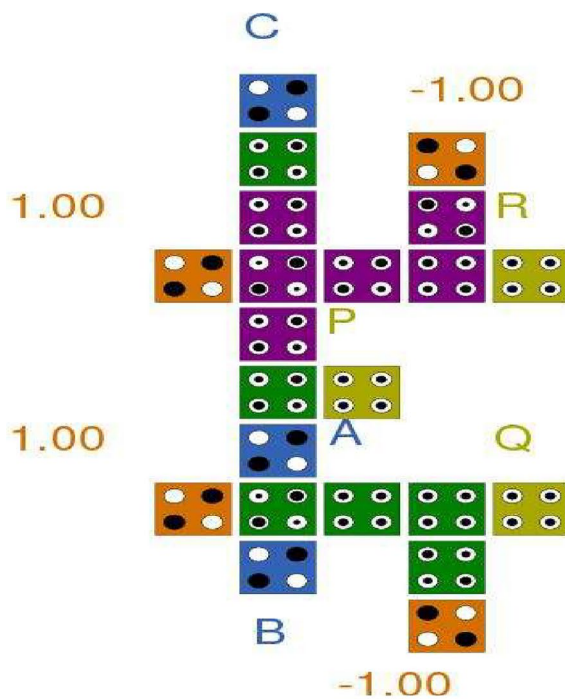
Fig. 2 Proposed XOR gate and its waveform

(a) Feynman gate-QCA

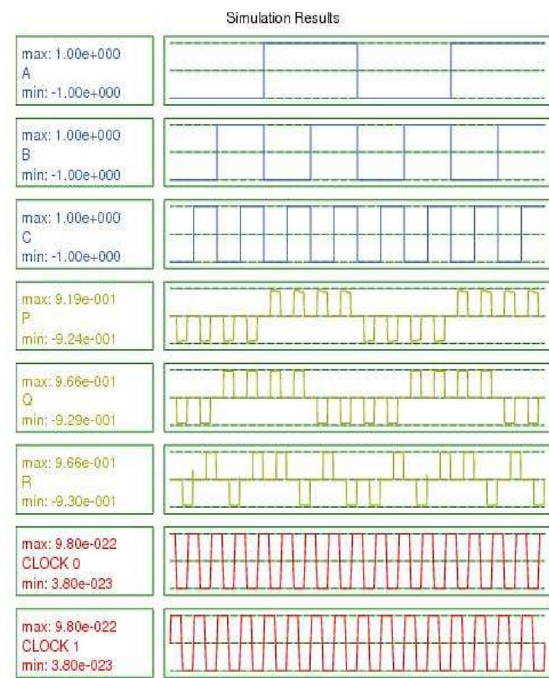


(b) Feynman gate output waveform

Fig. 3 Proposed Feynman gate and its waveform



(a) Double Feynman gate-QCA



(b) Double Feynman gate output waveform

Fig. 4 Proposed Double Feynman gate and its waveform

$$E = \frac{\hbar}{2} \Gamma \cdot \lambda.$$

Power dissipation is

$$P_{\text{total}} = \frac{\hbar}{2} \left(\frac{d\Gamma}{dt} \cdot \lambda + \Gamma \cdot \frac{d\lambda}{dt} \right).$$

Energy dissipation per cycle is

$$E_{\text{diss}} = \frac{\hbar}{2} \int \Gamma \cdot \frac{d\lambda}{dt} dt.$$

3.7 Reversible logic function

Conventional logic is irreversible, losing information and dissipating heat. Reversible logic provides a bijective input–output mapping, minimizing energy loss and supporting quantum computing applications [5, 6].

3.8 Evaluation metrics

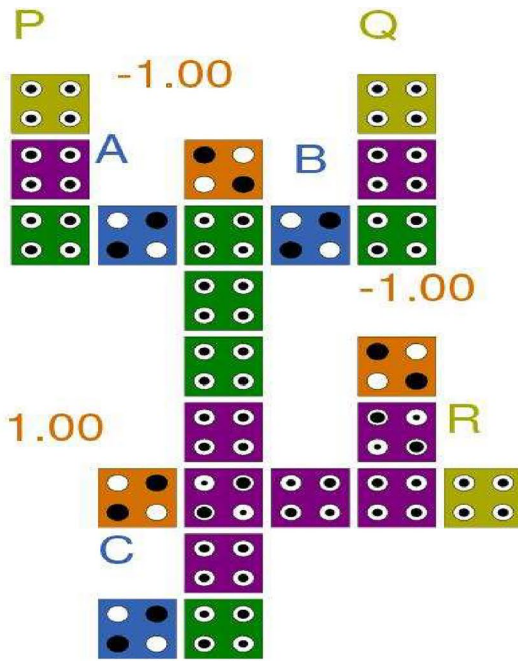
Evaluation metrics provide standards for assessing QCA circuit performance. Key metrics include complexity (number of cells), area (total cell footprint in μm^2), delay (maximum clock zones along the critical path, each contributing 0.25 units [8]), and cost (area $\times$ delay [26]).

Energy Analysis: Energy dissipation is evaluated per QCA cell, treating each as an energy reservoir with transfers occurring during clock cycles. Using QCADesigner-E, parameters such as $E_{\text{bath_total}}$ (energy to bath), $E_{\text{clk_total}}$ (energy transferred between cells), $E_{\text{error_total}}$ (sum of analysis errors), and $\text{Sum } E_{\text{bath}}$ (total simulation energy) are calculated [29].

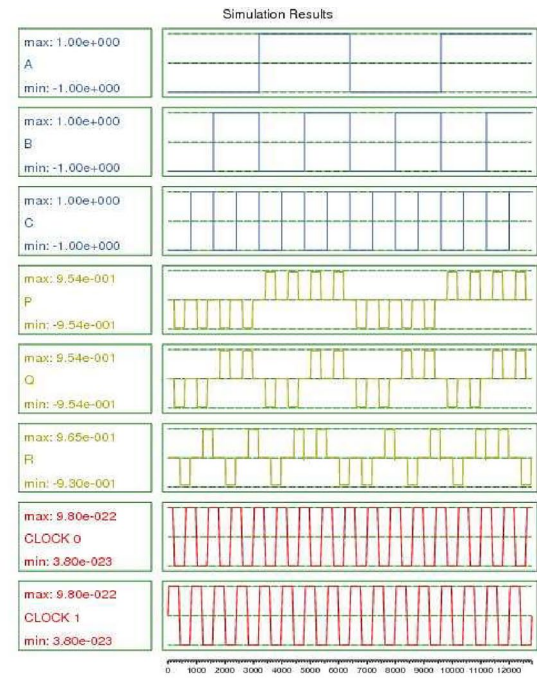
4 Proposed reversible gates

4.1 Reversible gates

Reversible logic gates serve as a basic component for sustainable computing, whose primary focus is reducing the consumption of energy and creating a null impact on the

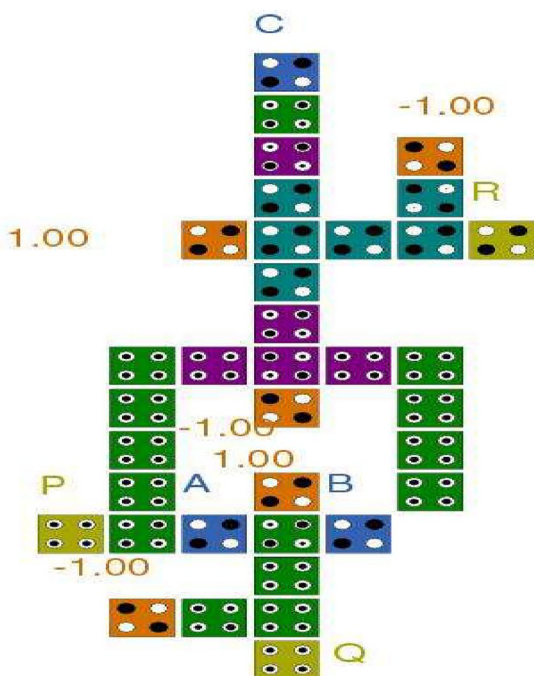


(a) Toffoli gate-QCA

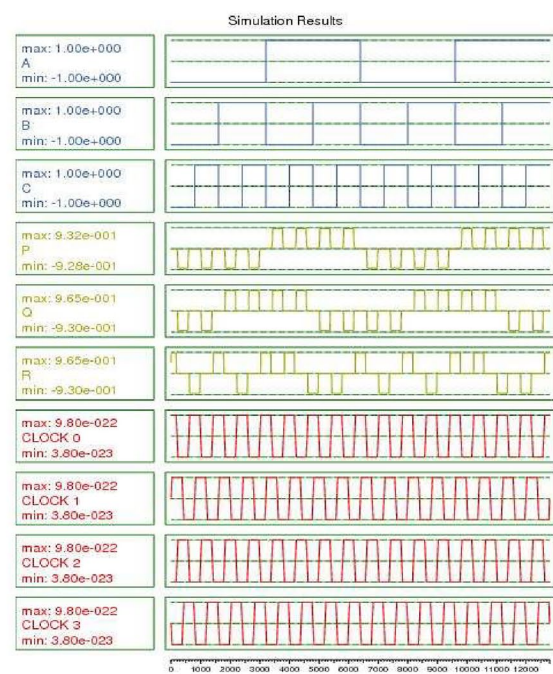


(b) Toffoli gate output waveform

Fig. 5 Proposed Toffoli gate and its waveform

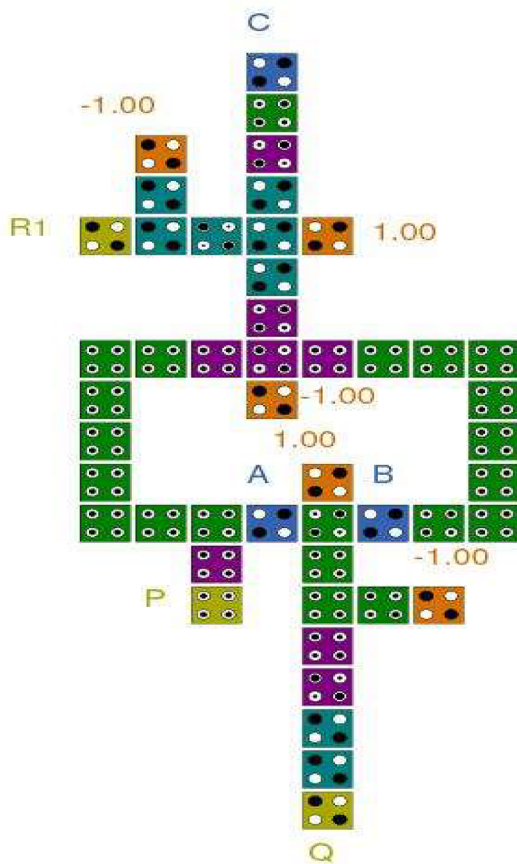


(a) TR gate-QCA

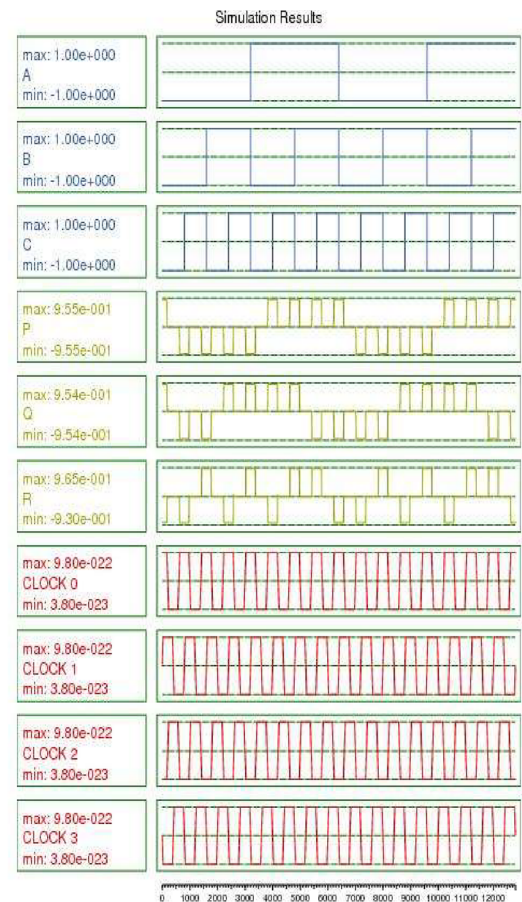


(b) TR gate output waveform

Fig. 6 Proposed Peres gate and its waveform



(a) Peres gate-QCA



(b) Peres gate output waveform

Fig. 7 Proposed TR gate and its waveform

environment. This work primarily focuses on the design of Feynman, Double Feynman, Peres, Toffoli, TR, and Fredkin reversible logic gates, as well as the design of circuits using these proposed reversible logic gates. The logic functions and the block diagram of these gates are shown in Fig. 1.

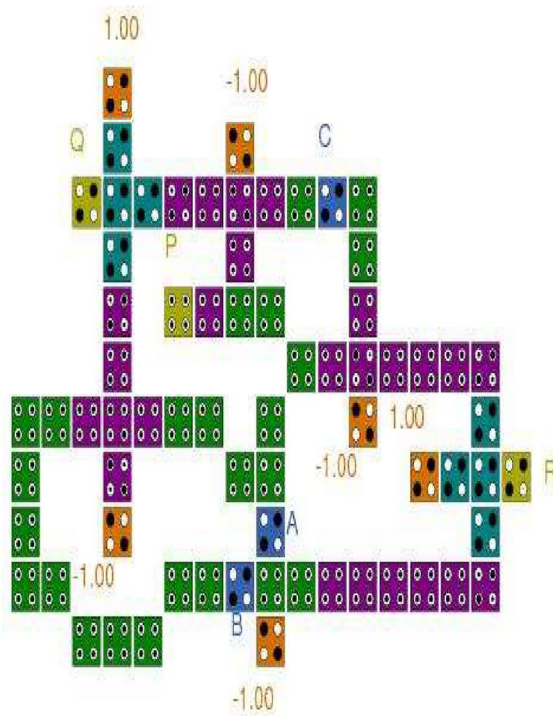
4.2 Proposed XOR gate

The XOR gate holds significant importance in the design process of reversible logic gates, serving as a foundational element in their design. The cell complexity and delay of reversible logic gates are directly impacted by the complexity of XOR gates. Prior research as [31–36], commonly employs the AND–OR–INVERTER method for designing XOR gate. Another approach, discussed in [26], introduces an XOR gate based on inter cellular effects, utilizing 14 cells.

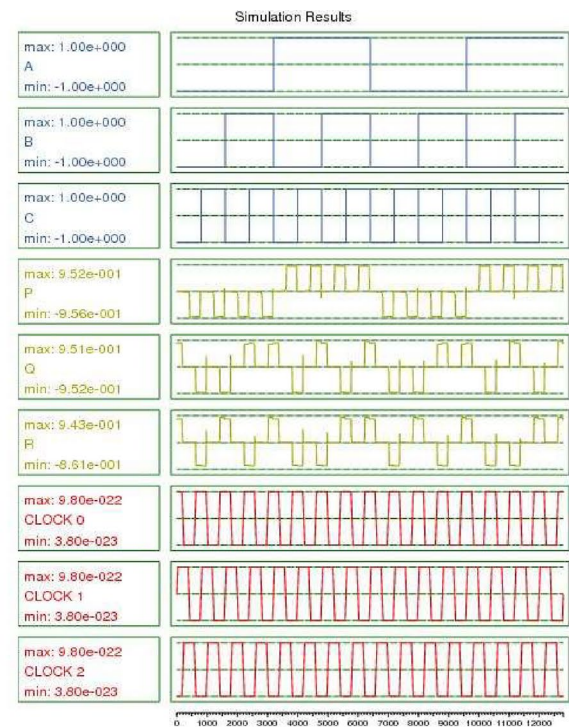
Many existing designs adopt multilayer architectures, leading to potential misalignment of quantum cells during fabrication, as noted in [14], where an XOR gate using 8 cells necessitates two clock cycles. However, increasing clock zones can result in higher delays, adversely affecting circuit performance. A novel strategy is proposed herein for crafting a low-power XOR gate, leveraging the intercellular effects method.

The proposed approach as shown in Fig. 2a is implemented based on the position of the cell rather than on the equation. The polarity of the cells is determined by the radius of influence. The proposed XOR gate utilizes only 9 cells, occupies an area of $0.003 \mu m^2$, and achieves a clock delay of 0.25. Notably, it is designed in a single layer, eliminating the need for cell rotation or wire crossing. In Fig. 2b, the output waveform is shown.

In addition, the authors [37] highlighted that QCA facilitates nanometer-scale, energy-efficient logic circuit design, making it highly suitable for digital comparators. Their proposed QCA-based reversible 1-bit comparator demonstrates



(a) Fredkin gate-QCA



(b) Fredkin gate output waveform

Fig. 8 Proposed TR gate and its waveform

substantial quantum cost savings and enhanced performance compared to traditional designs.

4.3 Proposed Feynman gate

Feynman gate, named after physicist Richard Feynman, consists of two inputs and two outputs. Input A is transferred to output P via wire, while Q is the XOR of both the inputs. Figure 3a depicts the proposed Feynman gate, which is designed using the proposed XOR gate. The proposed design uses 11 cells and occupies a space of $0.005 \mu\text{m}^2$, achieving a clock delay of 0.25 units with a quantum cost of 0.00125. In Fig. 3b, the output waveform is shown.

4.4 Proposed double Feynman gate

The double Feynman gate has two interconnected Feynman gates. Output P is derived from input A using a cell array. Output Q is produced by performing an XOR operation on A and B. Output R is derived by performing the XOR operation on A and C. The circuit is constructed with the proposed XOR gate. The proposed approach illustrated in Fig. 4a utilizes 22 cells, consumes $0.04 \mu\text{m}^2$, results in a clock delay

of 0.5 units, and incurs a quantum cost of 0.02. The output waveform is depicted in Fig. 4b.

4.5 Proposed Toffoli gate

The Toffoli gate combines a majority voter with an XOR gate. The majority voter operates an AND operation between inputs A and B while choosing the third input as "0." The output R is derived by XORing the result with input C. The outputs P and Q reflect the inputs A and B, respectively, using QCA wires. The XOR operation is carried out via the proposed XOR gate. The proposed Toffoli Gate illustrated in Fig. 5a utilizes 23 cells, occupies an area of $0.03 \mu\text{m}^2$, achieves a clock delay of 0.5, and has a quantum cost of 0.015. The output waveform is depicted in Fig. 5b.

4.6 Proposed Peres gate

The Peres gate consists of two proposed XOR gates and a single majority voter. Output P reflects input A, whereas output Q is derived from the XOR operation carried out by the proposed XOR gate on A and B. The output R is produced through the utilization of a majority voter together with a proposed XOR gate. The approach illustrated in Fig. 6a

Table 1 Comparative analysis of proposed design

Circuit	Author(s)	Cell complexity	Area (μm^2)	Delay (cycles)	Quantum cost (ADP)	Type of crossover
XOR Gate	Sasamal et al. [8]	30	0.020	0.75	0.015	Coplanar
	Niemier et al. [39]	60	0.090	1.5	0.135	Coplanar
	Hashemi et al. [40]	54	0.080	1.5	0.120	Coplanar
	Mohammadi et al. [41]	39	0.030	0.75	0.0225	Multilayer
	Singh et al. [42]	28	0.020	0.75	0.015	No crossovers
	Proposed	9	0.003	0.25	0.0075	No crossovers
Feynman Gate	Sasamal et al. [8]	26	0.030	0.5	0.015	No crossovers
	Roy et al. [9]	32	0.033	0.75	0.0165	No crossovers
	Kumar et al. [18]	58	0.064	1.5	0.032	No crossovers
	Debnath et al. [43]	90	0.112	1.75	0.056	Coplanar
	Mukherjee et al. [44]	34	0.025	0.75	0.0125	No crossovers
	Latha Gade et al. [45]	13	0.020	0.5	0.010	No crossovers
	Proposed	11	0.005	0.5	0.00125	No crossovers
	Sasamal et al. [8]	53	0.058	0.5	0.029	No crossovers
Double Feynman Gate	Abdullah-Al-Shafi et al. [32]	93	0.3013	0.75	0.225	Coplanar
	Seyedi et al. [46]	46	0.050	2.0	0.100	Coplanar
	Sasamal et al. [47]	40	0.045	0.5	0.0225	Coplanar
	Proposed	22	0.040	0.5	0.020	No crossovers
	Sasamal et al. [8]	59	0.060	0.75	0.045	Coplanar
	Das J.C et al. [48]	64	0.088	1.5	0.132	Coplanar
Toffoli Gate	Safoev et al. [49]	50	0.058	1.0	0.058	Coplanar
	Roy et al. [50]	40	0.050	2.0	0.100	Coplanar
	Proposed	23	0.030	0.5	0.015	No crossovers
	Roy et al. [9]	79	0.0835	0.75	0.6265	Coplanar
	Sasamal et al. [31]	273	0.570	6.0	3.420	Coplanar
	Proposed	48	0.070	0.75	0.0525	No crossovers
Peres Gate	Naz et al. [7]	69	0.079	1.0	0.079	Coplanar
	Roy et al. [9]	76	0.079	0.75	0.059	No crossover
	Das et al. [37]	122	0.090	1.0	0.090	No crossover
	Sarma et al. [51]	225	0.540	1.5	0.810	Coplanar
	Proposed	36	0.090	0.75	0.0675	No crossover

Table 1 (continued)

Circuit	Author(s)	Cell complexity	Area (μm^2)	Delay (cycles)	Quantum cost (ADP)	Type of crossover
Fredkin Gate	Sasamal et al. [8]	75	0.080	0.75	0.060	Coplanar
	Mohammadi et al. [52]	105	0.090	0.75	0.0675	Coplanar
	Latha Gade et al. [45]	75	0.080	1.5	0.120	No crossovers
	Kundu et al. [53]	80	0.140	1.5	0.210	Coplanar
	Proposed	69	0.100	0.75	0.075	No crossovers

utilizes 48 cells and occupies an area of $0.07 \mu\text{m}^2$, attaining a clock delay of 0.75 with a quantum cost of 0.0525. The output waveform is illustrated in Fig. 6b.

4.7 Thapliyal Ranganathan (TR) gate

The Thapliyal Ranganathan (TR), also known as the TR gate, resembles the Peres gate, with the distinction that an inverted B is provided as one of the inputs to the majority voter [38]. Input A is directed to output P through the use of wires. In between, output Q is produced through the proposed XOR gate, while R is obtained by using a majority voter in conjunction with the proposed XOR gate. The TR Gate presented in Fig. 7a employs 36 cells, covers an area of $0.09 \mu\text{m}^2$, attains a clock delay of 0.75, and has a quantum cost of 0.0675. The output waveform is illustrated in Fig. 7b.

4.8 Proposed Fredkin gate

The Fredkin gate illustrated in Fig. 8 consists of three inputs (A, B, and C) and three outputs (P, Q, and R). The input A is transmitted to the output P through a QCA wire, while output Q is represented as $A' B + A C$, and R is expressed as $A B + A' C$. Figure 8 illustrates the proposed design of the Fredkin gate in Quantum-dot Cellular Automata (QCA). The design can be simplified by using only majority voters, thus discarding the need for XOR gates. This modification reduces computational energy and significantly minimizes the required area.

4.9 Metric analysis of proposed gates

Table 1 presents a compilation of various parameters associated with different reversible logic gates.

The analysis evaluates the performance of the proposed design across many gates—XOR Gate, Feynman Gate,

Double Feynman Gate, Toffoli Gate, Peres Gate, TR Gate, and Fredkin Gate—relative to the optimal reference designs. Key performance indicators, including cell complexity, area, delay, and quantum cost, are assessed based on percentage enhancement.

The findings provide considerable enhancements for the suggested design in the majority of instances, with key highlights as follows:

The proposed XOR Gate attained a 67.86% enhancement in cell complexity and an 85.00% enhancement in area efficiency. Both delay and quantum cost resulted in a decrease of 66.67% and 50.00%, respectively.

The Feynman Gate exhibited exceptional performance, achieving a quantum cost reduction of 87.50%, an area enhancement of 75.00%, and a significant decrease in cell complexity of 15.38%.

The Double Feynman Gate demonstrated equitable enhancements, comprising 45.00% reduction in cell complexity and 11.11% decrease in area, without incurring delays or quantum cost penalties.

The Toffoli Gate demonstrated a 48.89% reduction in cell complexity and a 33.33% decrease in latency. Nonetheless, area and quantum cost measures exhibited negative values, suggesting possible trade-offs.

Peres Gate demonstrated moderate improvements, with 39.24% increase in cell complexity and 25.00% reduction in latency. The quantum cost improved somewhat by 6.36%, despite a trade-off in the area.

TR Gate attained constant advancements, with a 47.83% decrease in cell complexity, a 25.00% reduction in latency, and a 16.67% increase in quantum cost.

Fredkin Gate achieved a balanced performance with an 8.00% enhancement in cell complexity, 50.00% reduction in latency, and a 37.50% improvement in quantum cost.

4.10 Energy dissipation

The proposed gates exhibit varying average energy dissipation and minimal inaccuracies across different designs. The XOR gate shows an average energy dispersion of 0.324 meV with a minor deviation of 3.27×10^{-5} eV, while the Feynman and Double Feynman gates demonstrate average dissipation of 0.160 meV and 0.960 meV with small errors of -1.68×10^{-5} eV and -9.56×10^{-5} eV, respectively. The Toffoli, Peres, and TR gates exhibit dissipation of 1.06 meV, 1.48 meV, and 1.13 meV with corresponding inaccuracies of -1.06×10^{-4} eV, -1.34×10^{-4} eV, and -1.06×10^{-4} eV. Additionally, the Fredkin Gate configuration records the highest dispersion of 2.61 meV with a minor error of -2.47×10^{-4} eV, indicating consistent low-energy dissipation across all proposed designs.

5 Proposed one-bit comparator

The proposed one-bit comparator designed with proposed Feynman and proposed Toffoli gates represents a novel approach to digital circuitry. Unlike conventional irreversible comparators, reversible comparator conserve information, making them valuable for low-power computing and quantum computation. A one-bit comparator analyzes two logic states, A and B, and produces three possible output signals:

$$A > B \Rightarrow AB'.$$

$$A = B \Rightarrow A \oplus B.$$

$$A < B \Rightarrow A'B.$$

The block diagram of the proposed one-bit comparator is depicted in Fig. 9. The Feynman gate provides either the standard or inverted inputs to the Toffoli gate, based on the logical state of B. Specifically:

- If B is in logic-0, the Feynman gate provides the standard input A.
- If B is in logic-1, the Feynman gate generates the inversion of input A.

To derive the logic for $A = B$, the negation of the output Q' is considered. The Toffoli gate receives the negated output of P from the Feynman gate and is supplied with input C set to 0. The resulting output R1 of the Toffoli gate represents the logic for $A < B$.

Next, the output P_1 is inverted, and Q_1 is fed into another Toffoli gate. The resulting output R_2 represents the logic for $A > B$. In this context, P_2 and Q_2 are considered as garbage values.

Figure 10a displays the proposed one-bit comparator in QCA utilizing 63 cells. It occupies an area of $0.12 \mu^2$ and experiences a delay of 0.5, with a quantum cost of 0.06. Figure 10b presents the output waveform.

Observations from Table 3 and Fig. 11, revealed that the proposed design outperforms existing comparator circuits regarding cell complexity, size, latency, and quantum cost, showing its superiority for quantum computing applications. It obtains the minimal cell complexity of 63, significantly minimizing design complexity in comparison with existing designs, which vary from 87 to 319 cells. The proposed design takes an area of just $0.12 \mu m^2$, providing it the most compact alternative among the compared circuits.

Furthermore, the suggested design shows a delay of 0.5 clock cycles, aligning with the optimal performance of existing designs through the utilization of efficient Feynman and Toffoli gates. This insightful choice of gates results in a significant decrease in quantum cost, attaining an area-delay product of 0.06, which is considerably lower than any existing designs, including the second-best design with a cost of 0.055. The advancements illustrate the efficacy of the proposed design in reducing resource consumption while maintaining performance, rendering it an ideal option for quantum comparator circuits in resource-limited settings [37].

As shown in Fig. 11, the proposed design exhibits significant improvements in terms of cell complexity, area, and quantum cost when compared to several existing designs, particularly [30] and [33]. It reduces cell complexity by over 60% and quantum cost by over 90% in comparison to these designs, indicating a much more efficient and streamlined architecture. The area is also significantly reduced, with nearly 50% improvement over [30] and almost 67% over [33]. These reductions suggest that the proposed design offers a compact and highly optimized solution while maintaining or even enhancing performance.

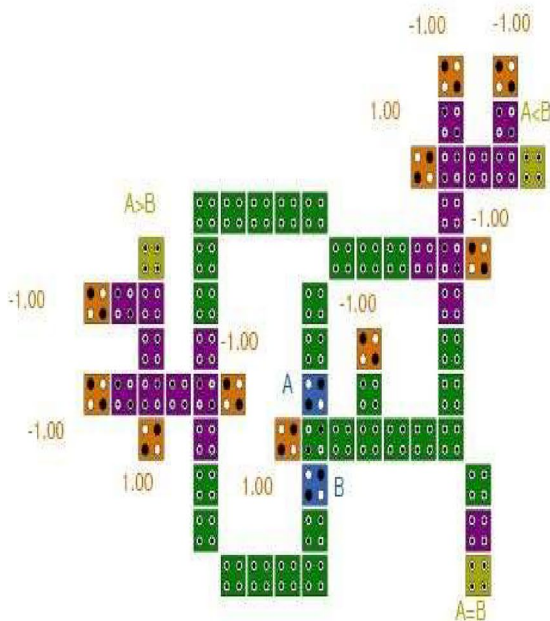
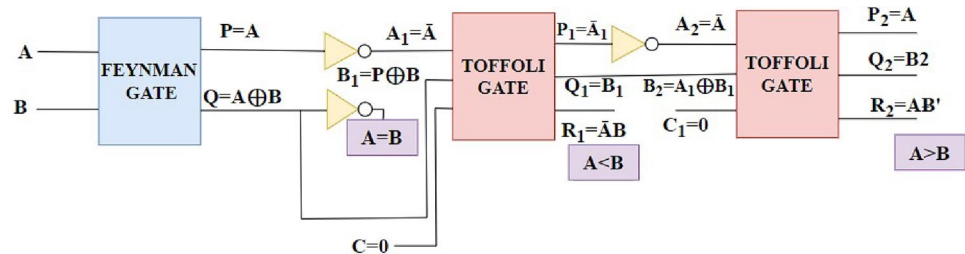
However, when compared to [37], the proposed system shows less dramatic gains. While it still improves cell complexity by about 27.6%, the area and quantum cost slightly underperform, with a 9.08% disadvantage in both metrics. This indicates that while the proposed system is highly efficient in most cases, [37] remains more optimized in terms of space and the area-delay product. Overall, the proposed design excels in simplifying and optimizing quantum circuits but may have trade-offs in specific scenarios.

Table 2 Average energy dissipation and inaccuracies of proposed gates

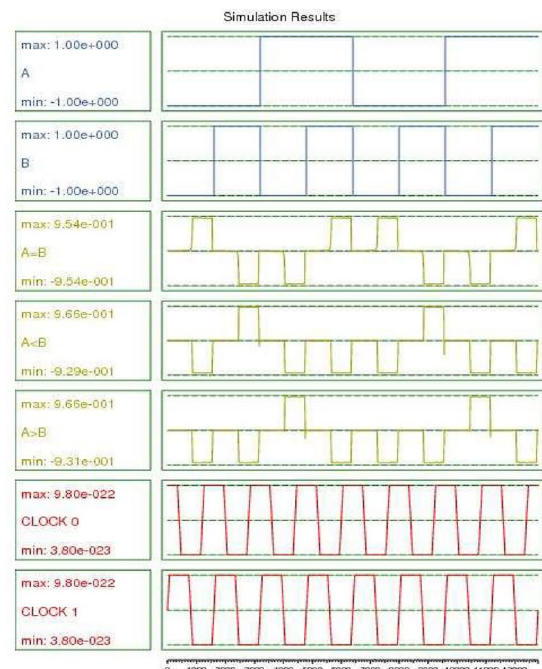
Gate	Average energy dissipation (meV)	Inaccuracy (eV)
XOR	0.324	3.27×10^{-5}
Feynman	0.160	-1.68×10^{-5}
Double Feynman	0.960	-9.56×10^{-5}
Toffoli	1.06	-1.06×10^{-4}
Peres	1.48	-1.34×10^{-4}
TR	1.13	-1.06×10^{-4}
Fredkin Gate	2.61	-2.47×10^{-4}

6 Proposed arithmetic and logical unit

The proposed arithmetic logic unit (ALU), shown in Fig. 12, and Fig. 13, features versatile functionality, capable of executing eight distinct arithmetic and logical operations based on control inputs K_2 and C , as well as operands A and B as shown in Table 4. Utilizing a concise layout comprising two proposed Toffoli gates and two proposed Feynman gates, the ALU achieves a cell complexity of 102 and utilizes an area of $0.17 \mu\text{m}^2$. Particularly, it operates efficiently without crossovers and exhibits a clock delay of 3 units. The ALU's quantum cost is measured to be 0.51.

Fig. 9 Proposed one-bit comparator

(a) Proposed one bit comparator

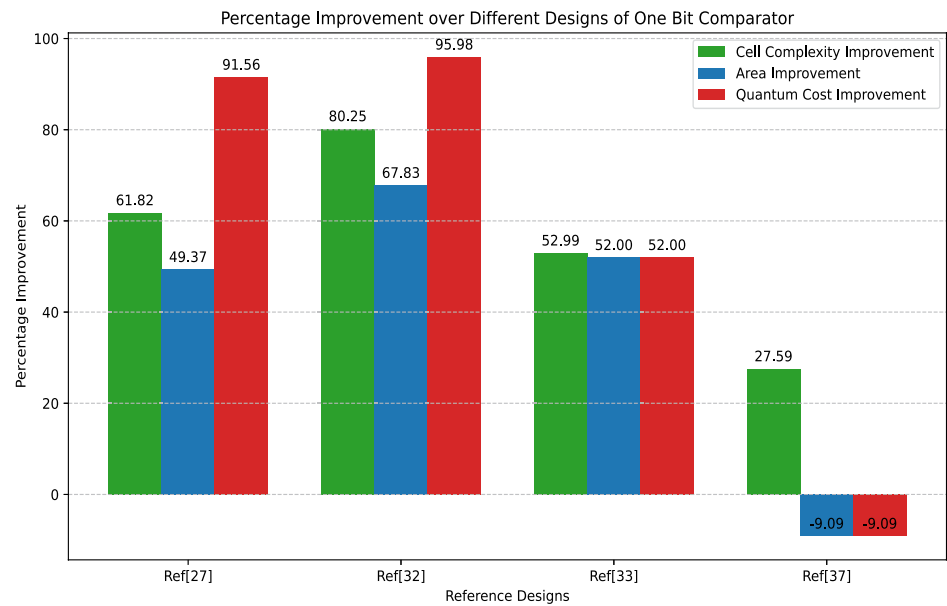
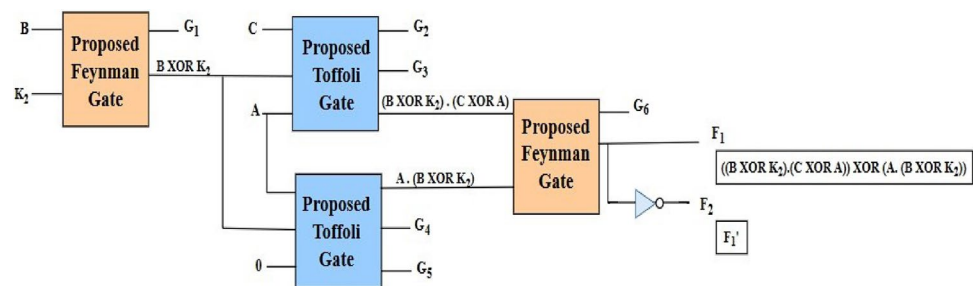


(b) Output waveform - one-bit proposed comparator

Fig. 10 Proposed one-bit comparator and its waveform

Table 3 Comparative analysis of comparator

Ref.design	Cell complexity	Area (μm^2)	Delay (clk cycles)	Gates used	Quantum cost (ADP)
Vahabi et al. [30]	165	0.237	3	TR and Feynman	0.711
Ahmed et al. [33]	319	0.373	4	Majority Gates	1.492
Akter et al. [34]	134	0.250	0.5	TR	0.125
Das et al.[37]	87	0.110	0.5	Feynman	0.055
Proposed	63	0.120	0.5	Feynman and Toffoli	0.060

Fig. 11 Percentage improvement of one-bit comparator**Fig. 12** Schematic of proposed ALU

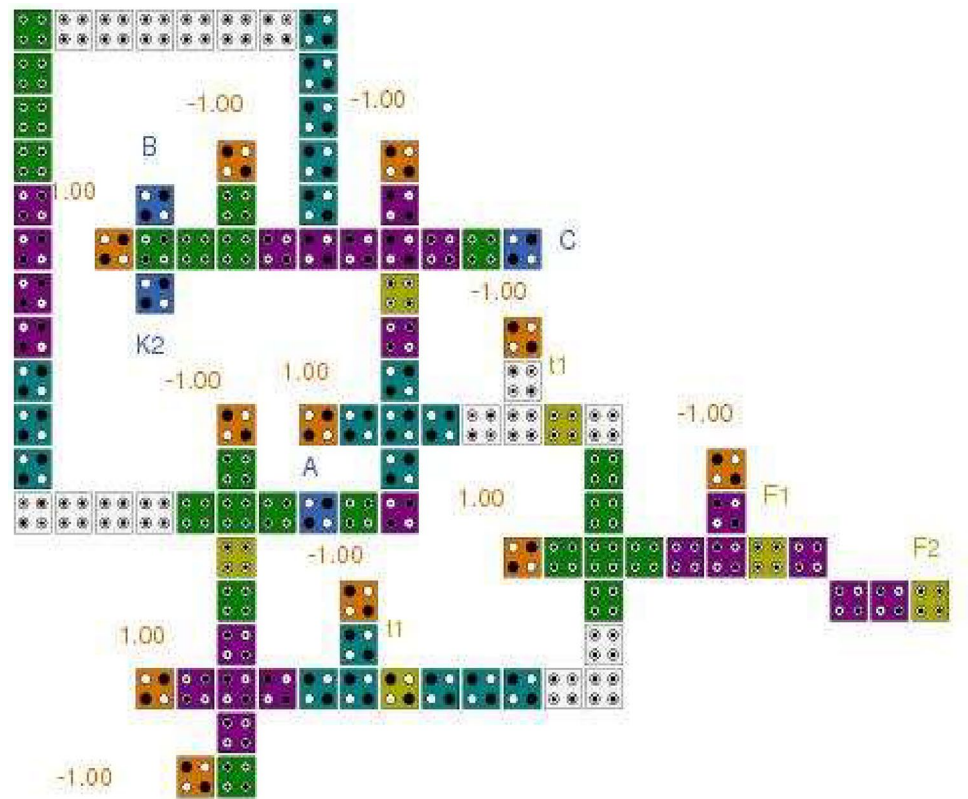
Its functionality encompasses essential operations such as transferring A , complementing A , performing logical operations such as OR, NOR, AND, NAND operations, and computing one's complement and two's complement subtraction.

The circuit produces outputs designated as G_1 to G_6 , F_1 , and F_2 , where G_1 to G_6 represent garbage outputs, and F_1 and F_2 correspond to the primary outputs of the ALU.

The equations governing the ALU functionality are as follows:

$$\begin{aligned} G_1 &= B, & G_2 &= C, & G_3 &= B \oplus K_2, & G_4 &= A, \\ G_5 &= B \oplus K_2, & G_6 &= CB \oplus K_2 \oplus A \end{aligned} \quad (1)$$

$$F_1 = ((B \oplus K_2) \cdot (C \oplus A) \oplus (A \cdot (B \oplus K_2))) \quad (2)$$

Fig. 13 QCA implementation of proposed ALU**Table 4** Comparative operation

Control inputs		Operation	
K_2	C	F_1	F_2
0	0	Transfer A	Complement of A
0	1	OR	NOR
1	0	AND	NAND
1	1	1's complement subtraction	2's complement subtraction

$$F_2 = (((B \oplus K_2) \cdot (C \oplus A) \oplus (A \cdot (B \oplus K_2))))' \quad (3)$$

6.1 Simulation waveform analysis

Figure 14, depicts the simulation waveform. In case I with control inputs K_2 and C set to 0 and operands A and B both equal to 0, the ALU performs the operation $A = 0$, resulting in output F_1 and F_2 being 1. In case II, where K_2 is 0, C is 1, and A & B are both 1, the ALU executes the operation $A + B = 1 + 1 = 1$, yielding F_1 as 1 and F_2 as 0.

Further in case III, with K_2 and Setting A to 1 and setting B & C to 0, the ALU conducts the operation $A \cdot B = 1 \cdot 0 = 0$, resulting in F_1 being 0 and F_2 being 1. Finally, in case

IV, where all inputs K_2 , C & A are 1, and B is 0, the ALU performs the operation $A + B' = 1$, resulting in F_1 being 1 and F_2 being 0.

These cases exemplify the ALU's versatility in executing various arithmetic and logical operations based on different combinations of input values.

The proposed ALU design shows notable improvements in all examined parameters, such as area, delay, and quantum cost, relative to the six reference designs, which is shown in Fig 15. The most significant enhancement can be found in the area, where the proposed design realizes a reduction exceeding 87% in comparison with [10]. This significant reduction shows the small size and physical efficiency of the design, which is essential for resource-constrained applications. Moderate improvements in the area are noted for [15] and [31], with reductions surpassing 82%, showing the proposed design's enhanced capability to reduce physical implementation needs (Fig. 15).

The proposed design shows a significant improvement in delay for the majority of reference designs, especially in comparison with [31], achieving a 40% reduction in clock cycles. This highlights the design's ability for high-speed operations, rendering it appropriate for real-time or latency-sensitive applications. The delay values in references such as [11] and [14] have been optimized and are near the proposed design, leading to minimal improvement in these instances.

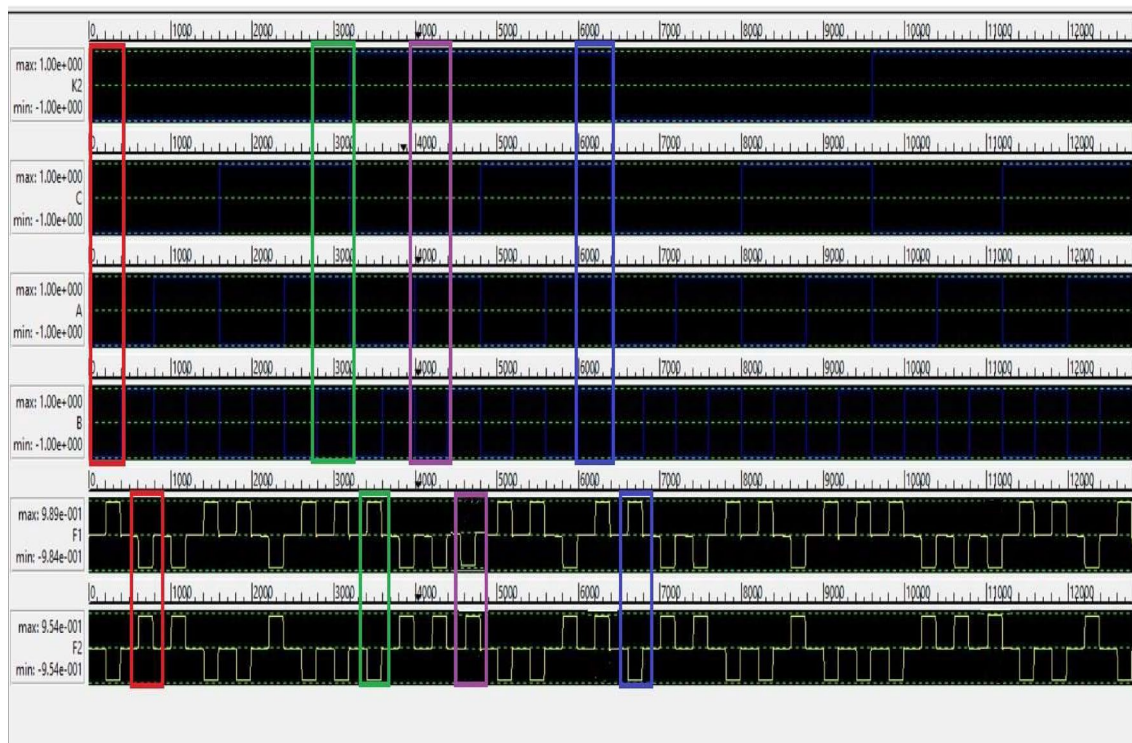


Fig. 14 Simulation result of proposed ALU

Table 5 Comparison of reference design and proposed ALU

Reference design	Complexity	Area (μm^2)	Delay (clk cycles)	Type of crossover	Number of operations	Quantum cost (ADP)
Ahmadpour et al. [10]	625	1.34	3.75	Coplanar	4	5.025
Oskouei et al. [11]	332	0.38	3	Multilayer	4	1.14
Chaves et al. [12]	1097	—	3.75	Multilayer	8	3.67
Rahimpour Gadim et al. [14]	464	0.78	3	Multilayer	4	2.34
Roy et al. [15] [15]	230	0.96	3.75	Multilayer	8	3.6
Sasamal et al. [31]	596	0.96	5	Multilayer	4	4.8
Proposed Design	102	0.17	3	No crossover	8	0.51

The quantum cost, integrating area and delay effects, indicates the overall efficiency of the proposed design. The findings show an improvement exceeding 89% compared to [10], highlighting the proposed design's efficacy in achieving a balance between compactness and performance. Consistent reductions in quantum cost are noted across all references, including [11] and [12], further supporting the proposed design's superiority in resource utilization and computational efficiency.

The proposed ALU design shows better efficiency compared to all references, particularly in quantum cost and area metrics, while also exhibiting competitive delay performance. The results indicate the design's robustness,

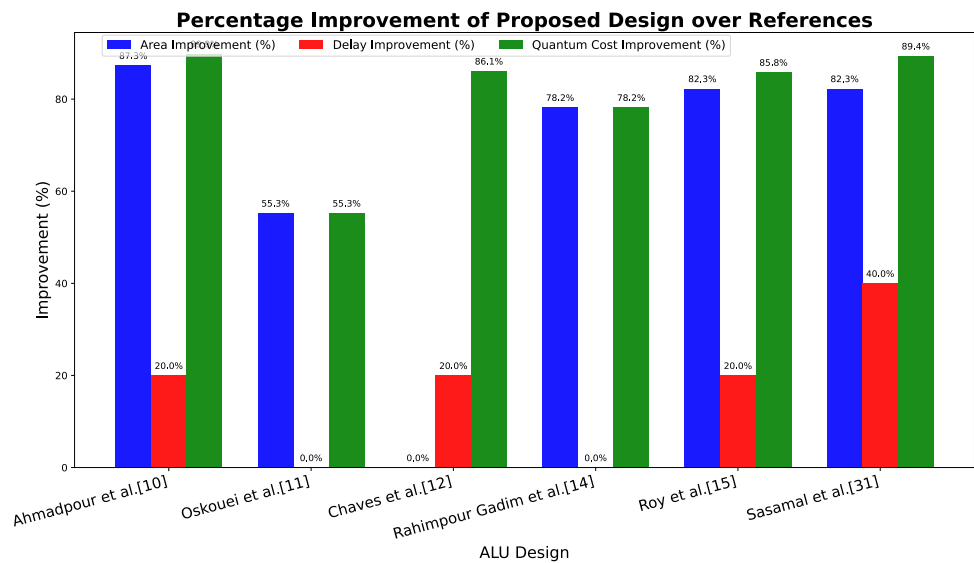
efficiency, and appropriateness for advanced quantum computing systems, positioning it as a viable solution for future applications. The proposed design effectively balances area reduction and delay optimization, making it a strong candidate for practical implementation.

7 Conclusion and future work

7.1 Conclusion

This study presents a comprehensive design and analysis of multiple reversible gates, including XOR, Feynman,

Fig. 15 Proposed ALU design—percentage improvement of Area, Delay and Quantum Cost



Double Feynman, Toffoli, Peres, TR, and Fredkin gates. The proposed gates demonstrate significant improvements in cell complexity, area, delay, quantum cost, and energy dissipation compared to existing designs. Notably, the second Toffoli configuration exhibits the highest energy dissipation of 2.61 meV with a minor error of -2.47×10^{-4} eV, indicating overall low-energy dissipation (Table 2).

Integration of these gates into practical circuits validates their effectiveness. The proposed one-bit comparator achieves minimal cell complexity of 63, an area of $0.12 \mu\text{m}^2$, a delay of 0.5 clock cycles, and a quantum cost of 0.06 (Table 3), outperforming existing designs by over 60% in cell complexity and 90% in quantum cost. The proposed ALU performs eight arithmetic and logical operations using two Feynman and two Toffoli gates, with a cell complexity of 102, an area of $0.17 \mu\text{m}^2$, a delay of 3 clock cycles, and a quantum cost of 0.51 (Table 5), exhibiting significant improvements in area, delay, and quantum cost over reference designs.

Overall, the results confirm that the proposed reversible gates and their integration into combinational circuits provide low-power, area-efficient, and high-performance solutions suitable for nanotechnology applications.

7.2 Future work

In the future, the QCA design model will have a lot of work to do, primarily on the design of memory element models like flip-flops and registers utilizing the suggested reversible architecture. These memory element designs will assist with the creation of low-power storage. Additionally, multi-bit and more intricate arithmetic circuits can be constructed and executed to evaluate the practical viability of the application.

Another key part of the reversible-based QCA design is the circuits that handle errors. Also, energy-aware synthesis methods can be changed to discover the least amount of energy loss while enhancing the area and delay parameters. Finally, these activities can be done to combine reversible logics to produce the hybrid system design architecture for the benefit of low-power applications in the part of computing architecture.

Author Contributions Dr. P. Rajasekar, Dr. H. Mangalam, and Mr. V. Shakhthivel worked on the study and analysis of the work model. Dr. P. Rajasekar and Mr. V. Shakhthivel took part in the literature survey and result execution. Dr. H. Mangalam and Dr. P. Rajasekar worked on results analysis and presentation. All authors contributed to the preparation and evaluation of the document.

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Declarations

Conflict of interest All authors have no conflict of interest.

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

Informed consent Not applicable.

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